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Individual Taste Preferences Predict Cortical Taste Dynamics but Are Modified by Experience

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eLife Assessment

This **valuable** study demonstrates how individual taste preferences shift over time, how these changes relate to cortical activity, and how experience reshapes both. The evidence is largely **solid**, although additional analyses are needed to strengthen some of the conclusions. The results should be of interest to neuroscientists studying sensory physiology.

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Abstract

As is true in humans, no two rodents prefer precisely the same tastes, and no one rodent shows precisely the same taste preference pattern forever. Here, we make use of this between- and within-animal variability to generate and test novel insights about the stability and malleability of neural perceptual coding. We used a brief-access task (BAT) to reveal both between-rat individual differences in taste preferences and shifting preferences within individual rats (quantified in terms of lick bout lengths). We moved on to show that these phenomena are not simply random variation: first, we demonstrated that gustatory cortical (GC) taste response dynamics, the late part of which have been shown to reflect palatability, match that individual rat's BAT preferences (evaluated almost 2 weeks prior) better than canonical preference patterns, revealing that the individual differences reflect differences in neural taste processing; this enhanced correspondence, however, links neural taste processing only to the most recent BAT session—GC palatability processing does not reflect performance in earlier BAT sessions. The finding that rats' perceptual preferences shift across BAT sessions carries with it the implication that any tasting experience might change these preferences. We tested this implication by performing a second session of GC taste-response recordings—sessions in which tastes were delivered by intra-oral cannula to passive rats—and found that palatability-epoch taste responses in this later session no longer matched the most recent pre-recording BAT session. Together, these data demonstrate that every tasting experience (regardless of the method of taste delivery) changes the rat's perceptual/neural processing of those tastes.

Introduction

The basic task facing an animal's perceptual systems—processing input such that the animal can generate appropriate behavioral output—is of particular importance when said input is gustatory, because taste-guided consumption decisions are made when a potentially dangerous substance is already in the mouth. Palatability (the hedonic value associated with a taste) is the primary

variable informing the selection of responses (swallowing or rejection) to such stimuli, and it is therefore unsurprising that a sweet (caloric) taste would be ubiquitously preferred, and a bitter (potentially toxic) taste ubiquitously shunned.

But despite the inescapability of the above logic, individual differences in palatability judgments abound. It is a given, in fact, that different human animals have different specific likes and dislikes when it comes to food; for any particular taste and flavor, there can be found people who love it and people who loathe it. And this is not just true for humans: rat taste preferences vary from strain to strain (Bachmanov et al., 1998 [↗](#); Grill & Bernstein, 1988 [↗](#)) and even from individual to individual (Bacharach & Calu, 2019 [↗](#); Inui-Yamamoto et al., 2017 [↗](#); Kotlus & Blizard, 1998 [↗](#); Loney et al., 2012). While part of the explanation for these individual differences is no doubt genetic (Armitage et al., 2025 [↗](#); Bachmanov et al., 2016 [↗](#); Blizard, 1999 [↗](#); Keskitalo et al., 2007 [↗](#)), a good deal of evidence supports the hypothesis that experience and eating history control individual taste preferences. Humans from different cultures and subcultures have notably different food preferences (Jamel et al., 1996 [↗](#)), and experience—both social and non-social—is known to impact taste preferences (Herman & Higgs, 2015 [↗](#); Robinson & Higgs, 2012 [↗](#); Story et al., 2002 [↗](#)). Social interactions can change the preferences of rodents as well (Galef & Wigmore, 1983 [↗](#); Posadas-Andrews & Roper, 1983 [↗](#)), and even innocuous experience has an impact on rat taste coding that could conceivably change preferences (Flores et al., 2016 [↗](#); Flores et al., 2018 [↗](#)).

The above work motivates the suggestion that neural taste processing should differ from animal to animal, and even from day to day within single animals, in concert with differences in preference patterns. But there has been very little work done on the relationship between individual differences and neural perceptual responses, and essentially none looking at the stability of this relationship through time. Here we perform this work, repeatedly evaluating rats' taste preferences using a brief-access task (BAT) that allows rigorous estimates of between-taste preference patterns in a single session, and comparing these perceptual behavioral data to assays of the rats' gustatory cortical (GC) responses to the same tastes—responses that have been reliably and repeatedly shown to reflect palatability in the “late response epoch” spanning the period between ~0.5s and ~1.5s following taste delivery (Katz et al., 2001 [↗](#); Mukherjee et al., 2019 [↗](#); Sadacca et al., 2016 [↗](#)).

Individual rats' BAT preferences proved to be a specific match for those rats' “Late-epoch” GC responses, despite: 1) 2-week “empty stretches” between collection of behavioral and electrophysiological data; 2) a range of sampled taste batteries; and 3) vast differences in taste administration methods (licking in the BAT rig; passive intra-oral cannula delivery in electrophysiology sessions) for the two types of sessions. But this match was delicate, in that even single tasting experiences changed perception—palatability-related GC responses matched the most recent BAT preference assessment but not assessments from earlier days, and a second test of the rats' GC taste responses one day after the first no longer matched the preference ranking from the most recent BAT preference test.

Together, these data demonstrate that: 1) palatability-related spiking in the late “epoch” of a rat's GC taste responses reflects that individual rat's taste preferences; 2) these preferences are relatively stable across multiple weeks, provided that the rat does not have intervening taste experience; and 3) the rat's perceptual preferences can quickly change with taste experience, regardless of whether the experience is self-acquired or experimenter-applied.

Results

Taste preferences vary between animals and across days

To measure each rat's taste preferences, we analyzed licking patterns from Brief-Access Task (BAT) data. This approach provides a rapid and reliable assessment of preferences across an entire taste battery; in particular, analysis of lick microstructure—most notably, calculation of the average size of lick clusters in 10-sec trials—has been shown to be an accurate assessment of the palatability of a proffered taste (Dwyer, 2009 [↗](#); Lin et al., 2012 [↗](#); Lin et al., 2025 [↗](#); Strickland et al., 2018 [↗](#)),

with smaller average cluster sizes equating to lower palatability. In the most rigorous test of preferences, comparison of cluster sizes provides an accurate picture of which tastes an animal likes more, and which tastes it likes less.

Figure 3A [1](#) shows average sucrose, NaCl, and quinine lick cluster sizes for two representative rats' initial BAT sessions. While both rats produced very small lick clusters when offered the bitter quinine, and both produced large lick clusters when offered NaCl, they differed notably in their sucrose cluster sizes. A 2-way ANOVA of these data revealed a significant rat $\times$ taste interaction ($F(2,65)=13.44$, $p < 0.001$), with one of the two rats preferring sucrose to NaCl ($p < 0.05$) and the other barely preferring sucrose to quinine ($p > 0.05$).

In even casual scrutiny of Figure 3A [2](#), which presents lick cluster sizes to every taste across all 9 rats (the red dashed box delineates the rats used for Figure 3A [1](#)), it is clear that while few rats didn't like sucrose, individual variability was the rule across the entire sample of rats, and that individual preference orders often diverged from the canonical palatability ranking (sucrose $>$ NaCl $>$ citric acid $>$ quinine). While we varied the precise battery offered from rat to rat (to ensure that results of our brain-behavior relationship tests were generalizable; see Methods), inspection of differences between sucrose and NaCl (typically the most palatable tastes) lick cluster sizes in every rat that sampled both (Figure 3A [3-top](#)) makes it clear that some rats prefer (this concentration of) NaCl over (this concentration of) sucrose, while others do not. Similarly, differences between citric acid and quinine (typically the most aversive tastes) lick cluster sizes (Figure 3A [3-bottom](#)) revealed individual differences in aversiveness: some rats disliked quinine more than citric acid whereas several did not. These impressions were statistically confirmed using one-sample t -tests against a null hypothesis of zero difference, with Bonferroni correction applied ($ps < 0.008$; see figure legends for exact t - and p -values).

Individual differences in taste preferences persisted across several days of testing, in that on the rats' final day of BAT testing (depending on the rat, this was either the 3rd or 5th session), the results continued to reveal robust individual differences in taste preference (Figures 3B [1-3](#); see figure legends for statistics and p -values, and note that one rat reliably shunned both quinine and citric acid after an initial lick bout, such that there was no error bar in the equivalent aversiveness of both tastes). These results support the suggestion that perceived palatability varies significantly from rat to rat, even after neophobia—a tendency to avoid novel tastes (Barnett, 1958 [1](#); Domjan, 1977 [1](#); Lin et al., 2012 [1](#)) that could possibly introduce rat-to-rat differences—has long since subsided.

Finally, a comparison of Figures 3A [3](#) and 3B [3](#) makes it clear that rats' preferences changed with testing: 2-way ANOVAs revealed that the relative individual preference for either sucrose or NaCl was frequently different in the first and last tests ($F(5,116)=40.28$, $p < 0.001$), as was the relative aversiveness of quinine and citric acid ($F(5,116)=18.34$, $p < 0.001$; in both panels of Figure 3B [3](#), “#” denotes significant differences between the 1st and last session for that particular rat). That is, rats' taste preferences changed between testing sessions.

Individual rats' most recent taste preferences are specifically reflected in late-epoch GC taste responses

If the between-rat differences measured in the BAT test veridically reflect individual differences in perceptual preferences (as opposed to representing random variability in behavioral expression of population-general underlying preferences), then they should be reflected in that rat's neural taste responses. More precisely, the fact that gustatory cortex (GC) activity encodes taste palatability in the late taste-response epoch (Jones et al., 2007 [1](#); Katz et al., 2001 [1](#); Mukherjee et al., 2019 [1](#); Sadacca et al., 2016 [1](#)) motivates the specific prediction that this activity should uniquely correlate with palatability rankings derived from each rat's own licking behavior (i.e., lick cluster size) during BAT testing.

To test this hypothesis, we recorded single-neuron spiking activity from GC in the same rats from which we collected preference data, while delivering tastants through pre-implanted intraoral cannulae (IOCs). A total of 230 GC neurons were recorded— 25.5 ± 12.1 /rat. Figure 4A [1](#) shows peri-

Figure 1. Brief-access rigs and experimental timeline.

(A) The brief-access rig—multiple bottles, each made available periodically. To the right are example data schematizing the analysis of rhythmic licking to pull out lick numbers and “clusters,” both of which reflect palatability. (B) The experimental timeline, showing preference testing sessions and the delay (for surgery and recovery) between the final BAT day and Electrophysiology recording/passive tastant deliveries.

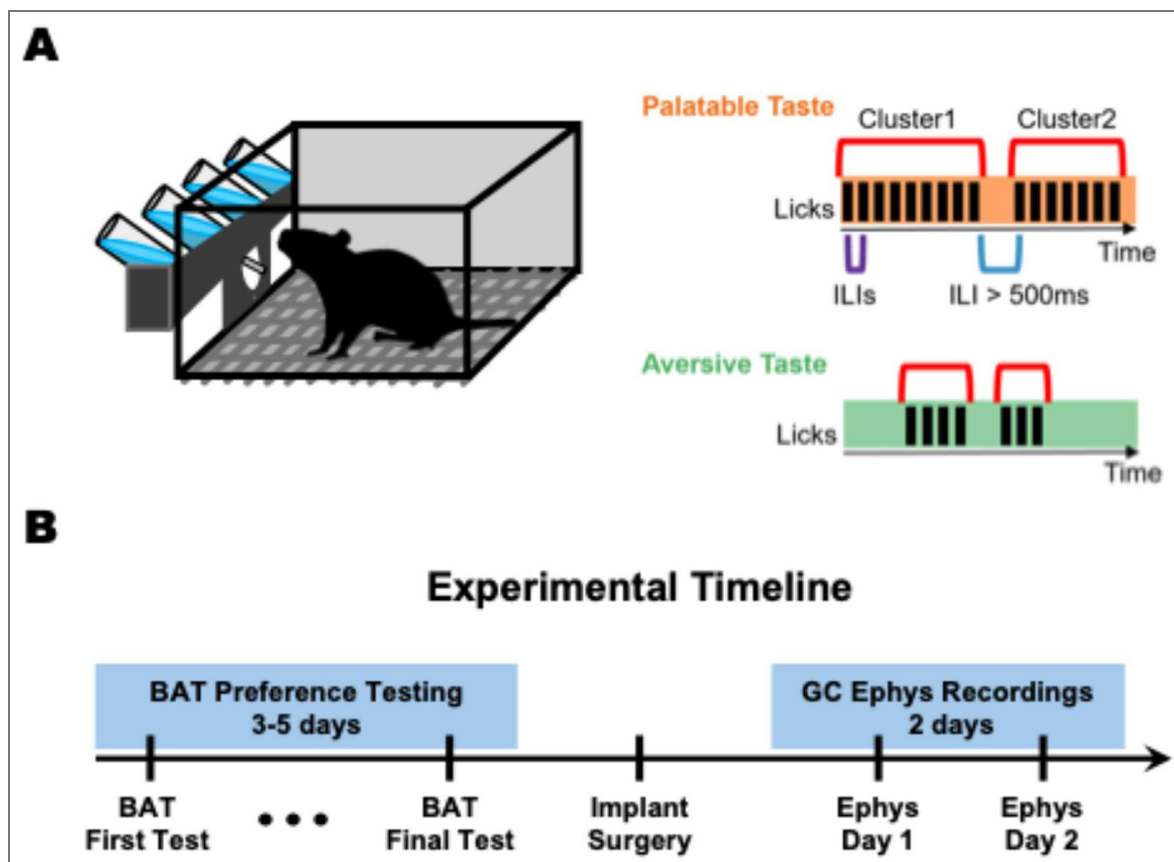
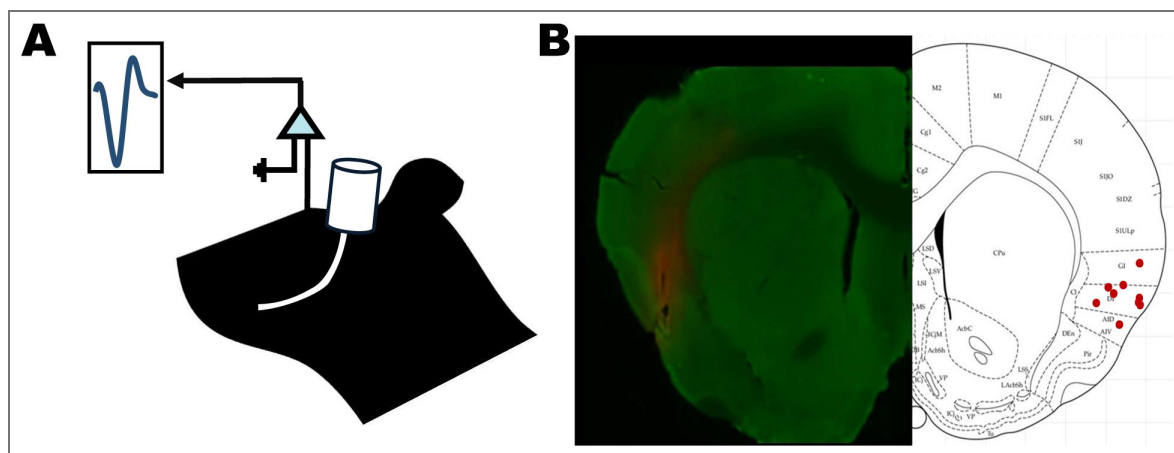


Figure 2. Passive taste delivery and electrophysiology recording and histological verification.

(A) A schematic of *in vivo* GC recordings with taste delivery *via* a pre-implanted intra-oral cannula. (B) Histological verification of a representative electrode implant site (left) with dye marking the recording site in gustatory cortex. On the right is the schematic of the coronal slice through central GC, with the positions of electrode tips for all animals (N=9) denoted with red circles.



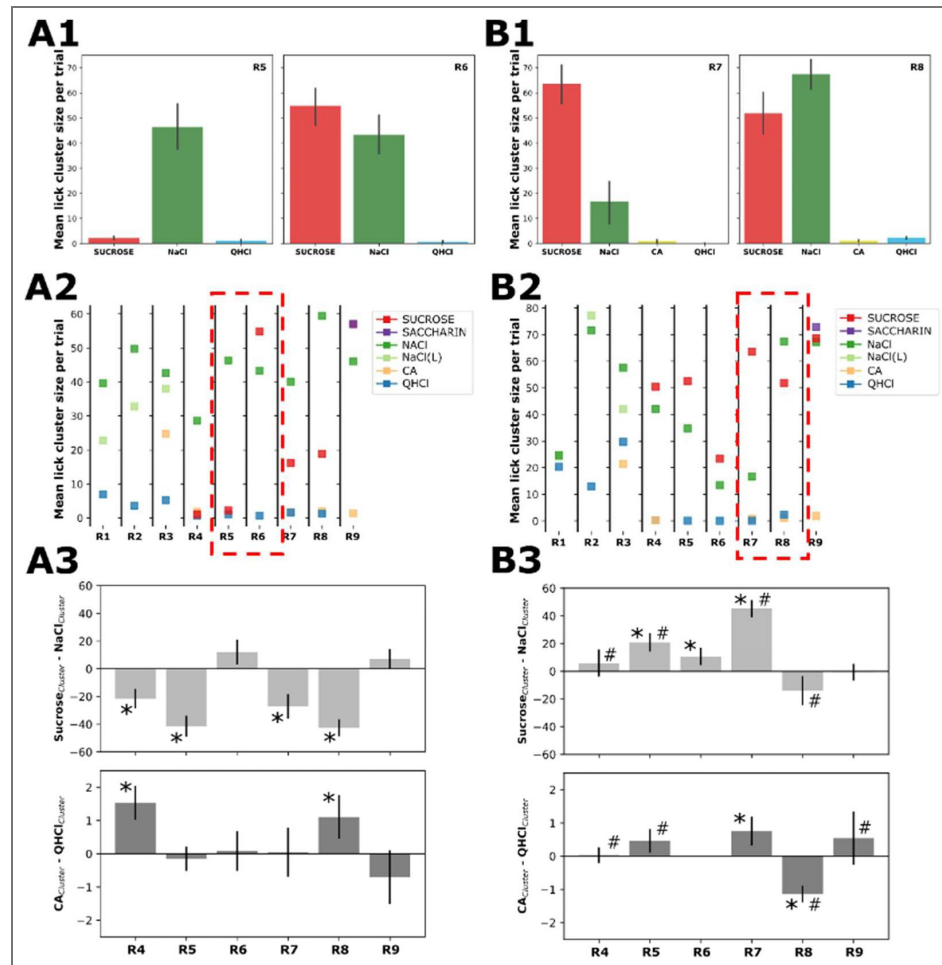


Figure 3. Individual and between-session differences in perceived taste palatability.

(A1) Mean lick cluster sizes for two representative rats during the first BAT test illustrate clear individual differences in preference for sucrose, NaCl, and QHCl. A two-way ANOVA revealed a significant Taste $\times$ Animal interaction ($F(2,65) = 13.44$, $p < 0.05$). Post hoc Tukey tests further showed that sucrose, while highly preferred by the rat on the left, was perceived similarly to QHCl by the rat on the right ($p < 0.05$). (A2) Mean lick cluster sizes for each taste across all animals during the first BAT test ($N = 9$). The two rats shown in panel (A1) are outlined for reference. (A3) Differences in lick cluster sizes comparing sucrose and NaCl (top) and comparing citric acid (CA) and quinine (QHCl; bottom) during the first BAT test. One-sample t -tests (against a population mean of zero) revealed individual differences in the preferences for the two palatable and the two aversive tastes ($p < 0.05$). (B1–B3) The same analyses as in A1–A3, but for each rat's final BAT session (depending on the rat, sessions 3 or 5). In (B3), we also show an analysis of whether individual rats changed their relative preferences between BAT sessions—two-way ANOVAs with variables Animal and Session, comparing the data shown in panels (A3) and (B3). The majority of rats exhibited significant changes in relative preference for tastes of the same valence across sessions ($\# p < 0.05$). The labels R1–R9 denote the rats included in the data analysis.

stimulus time histograms (PSTHs) for a representative GC neuron's responses to the same tastes delivered in that rat's BAT sessions, overlain on one another. These responses replicate those observed in many previous investigations (Katz et al., 2001 [↗](#); Mukherjee et al., 2019 [↗](#); Sadacca et al., 2016 [↗](#)): the firing caused by taste administration had little taste-specificity initially, becoming taste-specific only ~250 ms after delivery; the responses then came to reflect palatability only after the appearance of taste specificity, as indicated by the late appearance of significant correlations between firing rates and palatability after 500 ms of post-delivery time had elapsed (Figure 4B [↗](#)). Palatability-relatedness peaked (again consistent with previous reports) at approximately 1 second after taste administration, reaching a correlation of ~0.4 (i.e., $r^2=0.16$).

Note, however, that the correlation calculated from comparison of neuron firing to the “canonical” palatability rankings (green line) was, while significant, dwarfed by the much stronger correlations between GC late-epoch firing and palatability rankings estimated from the animal's own lick cluster sizes (NaCl > low NaCl > quinine > citric acid; Figure 4B [↗](#) inset) measured in the most recent BAT (orange line). That is, this rat's GC taste responses were specifically related to its own preferences. This neural expression of individually specific response tuning is particularly striking given the fact that: 1) the BAT preference evaluation took place almost 2 weeks prior to the electrophysiology session; and 2) the method of taste administration was starkly different in the two sessions—rats obtained their own taste samples through licking in the BAT, and passively received IOC deliveries of tastes in electrophysiology sessions. We can therefore conclude that individual differences in taste preferences as measured in the BAT are real, robust to these variables (see Discussion), and stable across weeks of recovery, at least when those weeks contained no experience with strong basic tastes.

In contrast, sessions of taste experience had a significant impact on preferences. Closer inspection revealed that only the most recent BAT preferences were reflected in neural activity. Figure 5A [↗](#), which presents the analysis for the entire sample of rats ($N = 9$), confirms that the magnitude of the late-response palatability-relatedness of firing evaluated using individual rats' final BAT data (orange curve) is more than twice as strong than when using canonical palatabilities (green curve; note that the small overall magnitude of r^2 values has to do with the lack of data selection—the fact that we did not restrict the analysis to those neurons with late-epoch palatability-related responses). This match to the rats' own data is entirely lost, however, when the first BAT session is used (blue curve). The robustness of this difference is underscored by the fact that only 1-3 days separated these different BAT sessions—1 to 3 days of tasting was enough to eliminate the effect. In fact, the peak neural correlation with palatability was slightly (but significantly) higher when canonical palatabilities were used than when earlier BAT sessions were used (we have no explanation for this small effect, but see Discussion for speculative consideration). Clearly, the repeated BAT sessions significantly altered the rats' preferences.

Figure 5B [↗](#) summarizes these results in an analysis comparing correlations between GC firing and preference patterns evaluated in different manners. The comparison was done separately for “early” (200–700 ms, a period during which GC responses largely reflect taste identity) and “late” (700–1200 ms, the time during which GC responses are coming to reflect palatability) epoch responses. As expected, effects were minimal during the early epoch, because responses at that time point were not palatability-related—the slight but significant difference between individual preferences measured in the most recent BAT and canonical preferences almost certainly reflects the onset of the Late epoch, a conclusion supported by the lack of significance in a comparison of the effect using first and last BAT rankings ($F(1,128) = 2.21$, $p > 0.05$).

Between-condition differences were pronounced, however, during the late epoch, when GC activity is known to reflect palatability. The brain-behavior correlation was significantly enhanced when using final BAT rankings (compared to when using canonical rankings; $t(128) = 5.71$, $p < 0.001$), and significantly poorer when using first BAT rankings ($t(128) = -3.20$, $p < 0.01$). The difference between these two effects was (of course) significant ($F(1,128) = 30.31$, $p < 0.001$). Palatability-related firing, which is restricted to the later aspects of GC taste responses, uniquely reflects the most recent assay of a rat's individual taste preferences.

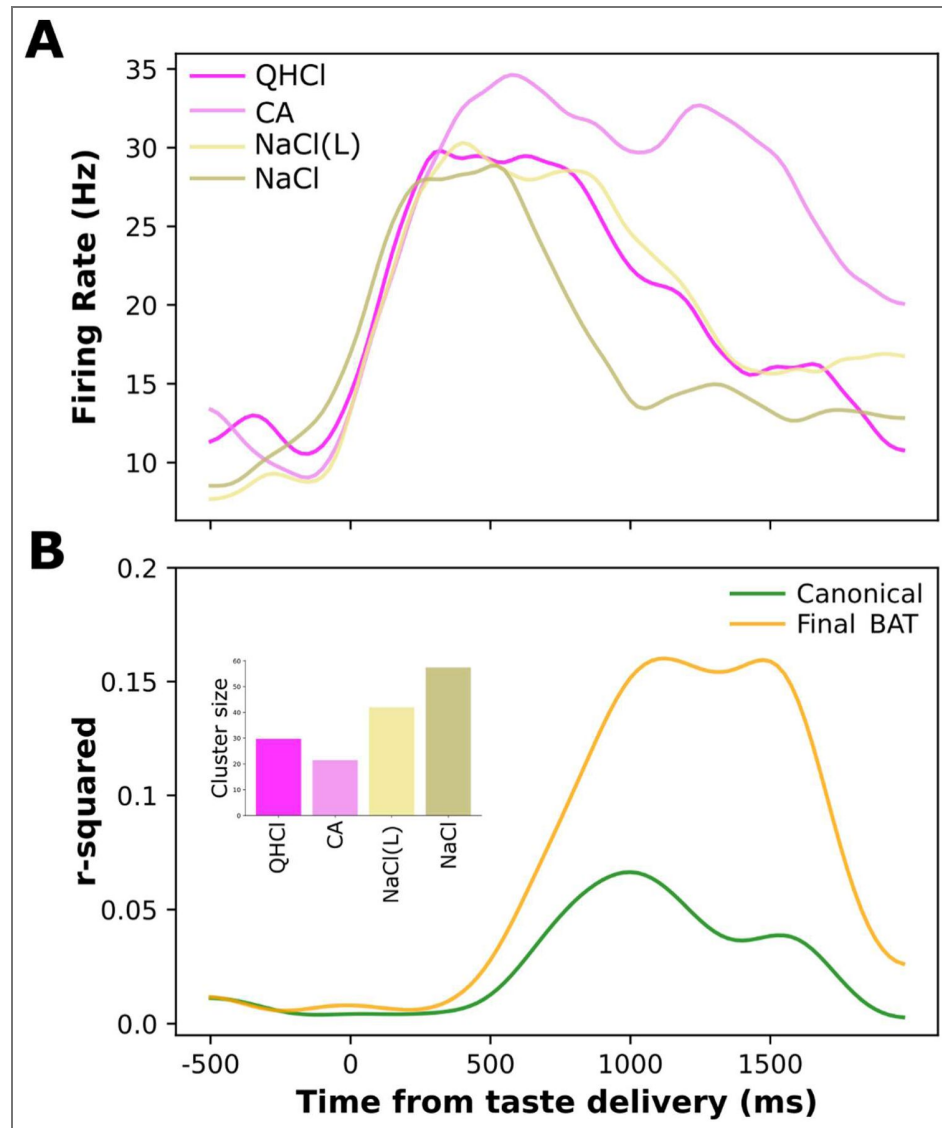


Figure 4. Palatability-relatedness of Late-epoch taste responses match the individuals' preferences.

(A) Peristimulus time histogram of 4 passively delivered tastes for a representative cortical neuron, showing typical response dynamics including correlation (r -squared, dotted line and right y-axis) to palatability peaking around 1 second from taste delivery. (B) Palatability correlations from the GC unit (A) calculated based on the canonical ranking or the ranks estimated from either first or final BAT test. **Inset:** Bar plot showing the average lick cluster sizes for each taste obtained during the final BAT testing session. Concentrations for each taste were as follows: QHCl, 7.8×10^{-5} M; CA (Citric Acid), 0.009 M; NaCl(L), 0.05 M; and NaCl, 0.1 M.

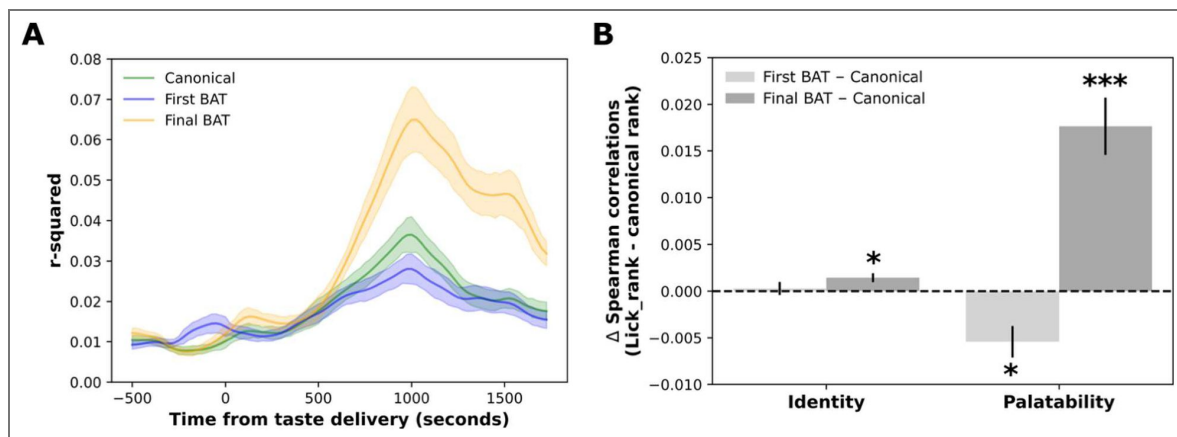


Figure 5. The correlation between Late-EPOCH GC firing and an individual rat's taste preferences is specific to the most recent BAT assay.

(A). The palatability correlation over time from all of the ($n=129$) recorded neurons, using canonical palatability (green), lick cluster data from the first BAT session (blue), and lick cluster data from the final BAT session (orange). **(B)** The difference in correlation calculated from canonical palatability and data from the animal's individual preferences (First BAT or Final BAT), separated into the Identity epoch (200-7000ms) and the Palatability epoch (700-1200ms). * $p < 0.01$, *** $p < 0.001$.

Any tasting experience changes rat taste preferences

The above results demonstrate that individual differences in taste preferences reflect between-rat variability in neural taste processing, but also demonstrate that taste experience in the BAT changes those preferences such that the match between an individual rat's palatability-related neural responses and earlier preference tests is lost. An obvious question regarding this result has to do with the specificity of the effect; it is important to ask whether it is only BAT-type taste experiences—experiences in which taste samples are acquired *via* licking, under the rat's control—that alter taste system function in such a way that preferences change. Alternatively, it is possible that any experience with strong concentrations of basic tastes will alter preferences (and neural processing).

This speculation engenders a specific novel hypothesis: if only self-administered experience with tastes licked from a source is sufficient to alter overall preference patterns, then electrophysiology sessions (which involve repeated administrations of each taste to passive rats *via* IOC) should have no such impact; if so, the palatability-related firing observed in a second electrophysiology session should continue to match the most recent prior BAT-based preference evaluation. We tested this novel hypothesis by conducting a second recording session 24 hours after the first to assess whether GC taste responses continued to correlate with licking-based individual preferences (note that between sessions, we advanced the chronically implanted electrodes by ~250 microns to ensure that the test was performed on a new set of neurons).

Figure 6 [↗](#) shows the result of this test. Figure 6A [↗](#), which presents the same moving-window assessment of correlations between neural firing and palatability used in Figures 4 [↗](#) and 5 [↗](#) (and in many previous publications), once again replicated the late-epoch rise in palatability-relatedness of GC activity; this late-epoch coding of palatability was robust across different specific measures of taste preference. The neural responses in this second electrophysiology test no longer reflected the rats' recently-assayed individual preferences, however, as evidenced by the fact that the correlations with canonical taste preferences, first BAT taste preferences, and final BAT taste preferences were essentially identical—there was no evidence of an enhancement of the GC response correlation with the most recent BAT data when using neurons recorded in the second electrophysiology session.

This observation was confirmed by analyses showing no significant difference between correlations with canonical palatability ranks and correlations with ranks derived from either the first or last BAT sessions (p s > 0.05; Figure 6B [↗](#)). We therefore conclude that both lick-acquired and IOC-delivered taste experience quickly changes taste preferences.

Discussion

The fact that our BAT data revealed individual rats to have distinct taste preference patterns (i.e., their own ideas of what tastes better and worse) is perhaps not surprising, although the magnitude of those individual differences is almost certainly larger than many would have predicted (particularly given the homogeneity of our rats' genetics and environment). More surprising is our finding that, even with a surgery, almost 2 weeks of recovery, and a major change in stimulus delivery method interpolated between the BAT and electrophysiological assays, individual behavioral differences are reflected in ("late-epoch") GC palatability-related spiking. Most surprising, however, are the perceptual changes wrought by even a single tasting experience—whether lick-acquired or IOC-delivered, taste delivery changed preference patterns. Apparently, every new tasting experience changes what a rat thinks of a taste in its mouth; at least as far as gustation is concerned, Heraclitus was correct that "no [animal] ever steps in the same river twice."

These results validate the use of a brief-access task (BAT), and more specifically of lick cluster size, to measure a rodent's palatability judgments, suggesting that the apparent individual and between-session variability that these tests reveal is *bona fide*. The significantly enhanced match between a specific rat's measured preferences and their GC late-epoch neural responses (compared to canonical average preference rankings) makes it clear that between-rat variability in

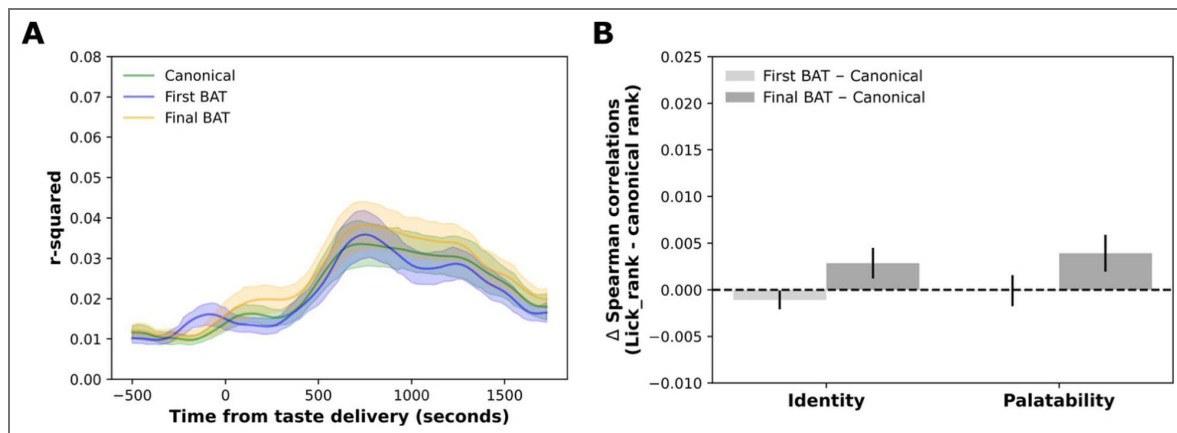


Figure 6. A single session of IOC taste delivery to passive rats nullifies the correlation with palatability calculated from the most recent BAT assay.

(A) Cortical coding no longer matches rats' individual preferences when using neurons ($n=101$) from electrophysiology session 2 (same conventions used in Figure 5A). **(B)** A summary of the data in **(A)**, showing there is no longer a significant difference using preference rankings estimated from BAT performance.

BAT performance is not noise but signal—each animal is genuinely different. This fact means that consumption averages calculated from BAT data should always have substantial confidence intervals around them. Furthermore, this brain-behavior match provides convergent evidence that the late-epoch GC response, which both predicts and drives the onset of consumption and rejection (Baas-Thomas et al., 2025 [↗](#); Li et al., 2016 [↗](#); Mukherjee et al., 2019 [↗](#); Sadacca et al., 2016 [↗](#)), is deeply relevant to behavior; the transition to palatability-related ensemble firing reflects the reaching of a consumption decision, made on the basis of neural processing that is to some degree specific to the animal and session.

This processing turns out to be remarkably stable (note that this means that the relationships between responses are stable—it does not necessarily mean that the precise responses are themselves unchanging) across days and weeks in which no specific experience with the basic tastes is provided. We performed electrode implantation surgeries after the last BAT preference testing session and gave each rat approximately 11 days to recover prior to the beginning of adaptation to the electrophysiology acquisition setup, such that almost 2 weeks separated behavioral and neural evaluations of taste palatability overall. Despite this separation, brain-behavior correlations were heavily influenced by the individual rat's BAT performance, a relationship that persisted despite the fact that BAT and electrophysiology sessions differed with regard to: 1) whether the rat was engaged in active sensing (BAT) or passive acceptance (electrophysiology); 2) whether the taste was licked (BAT) or delivered directly into the mouth *via* IOC (electrophysiology); and 3) whether the tastant accumulated in the mouth bit by bit (BAT) or in a single aliquot (electrophysiology). In essence, our experiments constituted a highly conservative test, in that each of the variables discussed above (including the 2-week between-session interval) would be expected to reduce the brain-behavior correlation; the fact that our results were nonetheless highly significant stands in testament to the robustness of our test, and the strength of our conclusions.

Regardless, the stability of preferences proves delicate, and dependent upon a lack of experience with sapid tastants between the BAT and recording sessions (note that in their home cages, rats may choose to partake only of bland chow and highly familiar biological detritus, and have no access to any strong exemplars of the basic tastes). Rats showed robust changes in their preferences across multiple BAT sessions, and only the preference pattern from the final BAT session matched the palatability-related GC firing recorded during the subsequent IOC session. This surprising result motivated a further test involving a comparison between two successive recording sessions: we reasoned that, if any tasting experience changes preference patterns, only the first recording session would be a good match for the individual differences measured in the prior BAT session; this test revealed that even tastes passively sampled through the IOC change perception; either of the two (very different; see above) types of taste experience is sufficient to change a rat's taste preference structure enough that it no longer matches earlier tests.

We consider this last point to be of particular importance. It is already well known that a first experience with a particular taste can increase the palatability of that novel taste—a phenomenon known as the attenuation of neophobia (Lin et al., 2012 [↗](#); Menchén-Márquez et al., 2026 [↗](#)). However, the phenomenon characterized in this current dataset is almost certainly not related to neophobia, in that: 1) these changes occur with each new session, even when the tastes can no longer be thought of as novel; and 2) these session-to-session changes cannot be described simply as an increase in each taste's palatability—if the palatabilities of all tastes increased together, preference ranks would have remained unchanged. Rather, our data suggest that aspects of taste processing that can otherwise remain stable across at least a 2-week period are altered by each and any taste experience, even after many taste experiences have already taken place.

By taking advantage of the naturally occurring individual differences in taste preference, the present findings demonstrate that both the stability and plasticity of these preferences are reflected in the aspect of GC activity that predicts and drives consummatory behavior. In the absence of taste stimulation, taste preferences remain largely stable. In contrast, whenever taste experience occurs, preferences change accordingly. This coordinated change between brain and behavior supports the idea that GC's role in taste processing extends beyond simple sensory

representation: it also contributes to the modulation of consummatory decisions by integrating sensory input with contextual and experiential information. In other words, GC functions not merely as a sensory encoder or decoder, but as an integrative hub that links perception with consumption behavior.

Materials and Methods

Subjects

Adult female Long Evans rats (N = 9; 250-375g at time of surgery) were acquired from Charles River Laboratories (Wilmington, MA), singly-housed in independently-ventilated cages on a 12h/12h light/dark schedule, and acclimated to the facility and handling for 1 week before the start of experimental procedures. Unless otherwise specified, animals in home cages had ad libitum access to lab chow and water. All procedures were conducted in accordance with the guidelines established by the Brandeis University Institutional Animal Care and Use Committee.

Experimental Overview

Rats' taste preferences were assessed in the Brief-Access Task (BAT) for 3-5 days ([Figure 1A](#)). After BAT testing, the rats were implanted with a 32-channel electrode into primary gustatory cortex (GC; [Figure 2AB](#)). Following recovery, rats received passive taste deliveries via IOC while GC activity was recorded for two sessions ([Figure 1C](#)). There were ~11 days total (surgery recovery and rig habituation) between the last BAT and first electrophysiological testing session.

Brief Access Task (BAT) Preference Testing

The BAT rig (Davis MS-160 Lickometer apparatus, Med-Associates Instruments) assessed individual preferences for each taste in the battery. Before preference testing, rats underwent three days of 45-minute habituation to the experimental chamber, after which they were placed on water restriction and given two days of licking training—60 10s-trials in which water was available via a metal lick spout attached to glass bottles.

For the preference test assessment, rats were presented with a battery of tastes (in each rat, one of 3 different possible batteries of tastes were used to ensure that results were generalizable—see [Table 1](#); as no differences in effects were observed between taste batteries, data was collapsed in our analyses) that spanned a putative range of palatabilities. One taste was offered per trial, in which rats were given 60s maximum to instigate licking, a maximum access time of 10s following the first lick, and a 30s inter-trial interval between trials. Tastes were presented in a pseudo-randomized order, such that two successive trials could neither offer identical tastants nor two aversive tastants (which would adversely impact willingness to participate). Each tastant was presented 8-10 times, depending on the specific taste battery. BAT testing persisted for 3 (taste set 1 & 2) or 5 (set 3) consecutive days.

Tastes, which were dissolved in deionized water (Millipore Milli-Q water purification system), were purchased from ThermoFisher Scientific at ACS Grade (with the exception of saccharin, which was FCC/USP Grade). A fan above the rig served to reduce any potential odor cues that might have accompanied taste presentation. The time of each lick was recorded for off-line preference analysis. All experiments were run between 10:00-16:00.

Surgical implantation of GC electrode and Intra-Oral Cannula

	BAT Sessions	Electrophysiology Sessions
Set 1	NaCl (0.1M) NaCl (0.05M) Citric Acid (0.009M) QHCl (7.8E-5M) Citric Acid (0.00625M) QHCl (0.00156M) Water	NaCl (0.1M) NaCl (0.05M) Citric Acid (0.009M) QHCl (7.8E-5M)
Set 2	Sucrose (0.3M) NaCl (0.1M) Citric Acid (0.1M) QHCl (0.001M) Water	Sucrose (0.3M) NaCl (0.1M) Citric Acid (0.1M) QHCl (0.001M)
Set 3	Sucrose (0.3M) Saccharin (0.005M) NaCl (0.1M) Citric Acid (0.1M) QHCl (0.001M) Water	Sucrose (0.3M) Saccharin (0.005M) NaCl (0.1M) Citric Acid (0.1M)

Table 1. Summary of taste stimuli used in brief-access tests (BATs) and electrophysiological recording sessions.

QHCl: quinine hydrochloride.

After the final BAT session, animals were removed from water deprivation and allowed to recover to full, pre-restriction weight for implantation surgery. Rats were anesthetized with an intraperitoneal injection of ketamine (100 mg/kg) and xylazine (5 mg/kg), and craniotomies were made to expose GC. The drivable electrode, consisting of a bundle of 32 nichrome wires (25 μ m in diameter), was implanted just above GC (A/P = +1.4, M/L = $\pm$ 5.0, D/V = -4.4), and secured in place with dental acrylic. Rats also received an intra-oral cannula (IOC)—a polyethylene cannula inserted behind the maxillary molars, through the left masseter muscle and through the opening of the scalp.

Rats recovered for 7 days; during the first day, meloxicam (Alloxate, 5mg/ml, Patterson) (1mg/kg) was injected for pain management.

Electrophysiological recording with passive tastant delivery

After recovery from surgery, rats habituated for 2 days (30-minute sessions) to the electrophysiological recording rig (plugged into their electrode harness in a plexiglass experimental chamber measuring 8.5 x 9.5 x 11.5 in), after which they were placed on water restriction in preparation for recording experiments. Experiments began with two days of habituation to liquids delivered through the IOC, during which 60 (day 1) and 120 (day 2) trials of water were delivered with inter-trial intervals of 22s. Twenty-four hours after, each rat received two consecutive recording days where 30 randomized presentations of each tastant (a total of four tastes selected from the same taste set received during the BAT preference testing; [Table 1](#)) were delivered.

Delivery of each taste was individually controlled by solenoid valves, pressurized by inert nitrogen. To ensure our ability to deliver precise amounts of fluid onto the tongue of each IOC delivery, we calibrated the rig to deliver 30 microliters in each aliquot at the beginning of each recording session, controlling the open time of valves *via* a Raspberry-Pi microcomputer.

Spiking data were collected *via* Intan RHD2132 analog-to-digital chip amplifier. Recordings were taken at a sampling rate of 30 kHz. Electrodes were driven down ($\sim$ 0.25mm) 24hr prior to each recording session to capture different populations of GC neurons for each session. The entire recording rig was enclosed in a faraday cage to reduce latent electromagnetic interference during recordings.

Electrophysiology data processing

Bandpass filtering and common average referencing were used to clean data and increase signal quality. Discriminable action potentials of no less than 3:1 signal-to-noise ratio were isolated online from each signal and saved digitally. Prior to analysis of taste-related activity, cells were clustered using 3-D cutting techniques alongside supervised verification of inter-spike interval plots. All spikes included in analysis had at least 2000 waveforms and met our criteria of < 0.5% 1ms violations and <2% 2ms inter-spike interval violations. For a comprehensive explanation of the protocol used for spike sorting, refer to Mukherjee et al. (2017). By driving down the mini-microdrive between sessions, new cells were captured each day, which was verified by comparing waveform shapes.

Semi-processed spiking arrays were then exported to an analysis framework built in python. After processing, each 7000ms (2s pre-stimulus, 5s post-stimulus) recording window was binned using a sliding window average with a window size of 250ms and a step size of 25ms.

Data analysis

Computational analysis of BAT data and preference ranks

Lick data was processed and analyzed in custom Python scripts. Libraries integral to the analysis included: pandas, numpy, pingouin, matplotlib, and seaborn. Several lick parameters were measured for each tastant presentation; we used the inter-lick interval (ILI; $\sim$ 125msec) to

determine lick cluster size—the number of licks produced prior to a 500ms pause (Figure 1A). Individual taste preferences were estimated using averaged lick cluster sizes obtained from a BAT session.

Palatability correlation

The method of calculating correlation between GC activity and palatability has been described previously (Katz et al., 2001; Sadacca et al., 2016): a vector of firing rates in each taste response at each 25-ms timepoint in a moving average is Spearman correlated (to control for nonhomogeneity) with the vector of preferences. For the “canonical” palatabilities, the data are first averaged across animals. To incorporate individual preferences here, we also correlated neural activity to animal-matched mean cluster values over all BAT sessions and in the first or most recent BAT session (First BAT, Final BAT).

Histology

Following the experimental sessions, subjects were deeply anesthetized with ketamine/xylazine mix (200, 20 mg/kg respectively, delivered *via* intraperitoneal injection) and perfused *via* intraventricular perfusion of saline, followed by 10% formalin. Brains were then extracted for histological verification of electrode placement. Electrode cannulae were painted with a fluorescent cell-labeling dye (Vybrant DiI, invitrogen) upon insertion to verify electrode location and regional markers were used to confirm the cannula track is above and the electrode tip sits in GC as confirmed by another independent experimenter. Following histological examination, one of the 10 experimental rats was removed from the experiment due to electrode misplacement, which left a total of 9 rats in the dataset (Figure 2).

Data availability

The data within this study can be made available from the corresponding author upon reasonable request. Spike sorting code used here as described above is available at https://github.com/narendramukherjee/blech_clust. This study’s design and analyses were not pre-registered.

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Peer reviews

Reviewer #1 (Public review):

Summary:

Maigler et al. set out to test the hypothesis that individual differences in taste preferences are (in part) due to individual differences in central taste processing. The first tested rats' preferences for a variety of taste stimuli on multiple days. They then recorded responses of neurons in the taste cortex to the same tastes on two consecutive days.

Strengths:

The authors collected high-resolution behavioral data from the same animals across multiple days, allowing for a detailed characterization of individual variation in taste preferences. They then performed recordings from the same set of animals in response to the same stimuli, allowing them to draw parallels between behavioral and neural responses. They convincingly show that preference ranks for a variety of basic tastes change over time and that the correlation between neural responses and preferences is not stable, correlating more strongly with more recent measures of preference.

Weaknesses:

Behavioral analysis: Data presentation does not show how preferences change over the course of testing. In particular, it is unclear whether there are any systematic changes in preferences over the course of testing that could explain the observed changes in correlation with neural responses, such as changes due to learning (e.g., flavor nutrient conditioning, relief of neophobia), changes in deprivation state, or habituation to/proficiency with the BAT setup. A secondary point is whether any changes in preference are attributed to internal individual versus external contextual factors. Both types of variation (i.e., across individuals and across time within an individual) are mentioned in the introduction, but it is not clear what the authors believe about the nature or neural representation of these sources of variation.

With respect to neural data analysis, no individual animal/day data are shown, making it difficult to assess the extent to which differences in correlation match individual differences in preferences and/or changes in preference with time within individuals. The correlation analysis is also lacking control for the fact that there is a certain degree of "chance" associated with behavioral and neural measures having matching ranks.

Finally, the conclusion that correlations between final day preferences and neural responses obtained from the second recording session are the result of experience needs more justification; it is unclear to what extent changes in correlation may be attributed to overall changes in responsiveness of the neural population.

<https://doi.org/10.7554/eLife.110155.1.sa3>

Reviewer #2 (Public review):

Summary:

The study from Maigler et al investigates how between- and within-animal differences in taste preference relate to differences in neural responsiveness. The experiments rely on an elegant combination of behavioral assays to measure preference (e.g., repeated brief access testing, BAT) and electrophysiological recordings to monitor the activity of ensembles of neurons in the gustatory cortex (GC) of rats.

BAT with distinct batteries of tastants revealed pronounced variability in preference (measured as licking bout size) across individuals. This variability across individuals persisted after repeated testing. Repeated BAT also revealed that each rat's preference for different tastants changed across time.

Electrophysiological responses of GC neurons to batteries of tastants showed that firing in the "late epoch" of taste processing (i.e., 500ms post taste delivery) correlated more strongly with the individualized rat's BAT preference rather than with a canonical preference ranking. Importantly, this correlation was stronger for the last BAT session compared to the first. Finally, the author shows that the correlation disappeared in a second, consecutive recording session, indicating that exposure to tastants reconfigures preferences.

Strengths:

- (1) The experimental design allows for an unprecedented look at the relationship between individual variability in taste preferences and neural processing.
- (2) The study demonstrates that taste preference variability is not mere experimental noise but reflects the dynamic nature of taste. A key strength is the clear evidence that behavioral variability is reflected in neural activity patterns, establishing a strong correlation between brain and behavior.
- (3) The evidence that simple exposure to familiar tastes can reconfigure preferences and taste representations is interesting.

Weaknesses:

- (1) The manuscript could use additional corollary analyses to provide a more complete picture of the phenomenon. For instance, how many neurons (per animal and in total) have significant correlations with the final BAT patterns? And with the first BAT? Can a time course of such counts be provided? Can some decoding analyses be performed at a single session level to reconstruct a rat's behavioral preference pattern from its neural activity?
- (2) The manuscript could benefit from additional polishing, both in the text as well as in the figures.

<https://doi.org/10.7554/eLife.110155.1.sa2>

Reviewer #3 (Public review):

Summary:

Maigler & Lin et al present a compelling set of behavioral and electrophysiological experiments exploring how individual differences in taste preference map onto neural responses in the gustatory cortex (GC). They go on to examine how both preferences and neural responses shift following intervening taste experience. Their experiments are strengthened by examining tastes of distinct identities and palatability (sweet, sour, salty, bitter) and corresponding each animal's individual preference to the palatability-related late phase of the neural response.

Strengths:

- (1) They demonstrate a relationship between the behavioral expression of taste preference and palatability-related GC neural responses. The direct correlation of expression of taste preference with GC neural responses indicates that taste preference behavior may be less noisy than previously thought, reflecting actual neural activity.
- (2) They address the stability of individual taste preference by comparing within and between session expression. This finding indicates that individual preference on any given test session can differ from canonical palatability.
- (3) They provide evidence that representational drift in palatability coding may arise from sensory experience rather than from the passive passage of time. The findings are novel and potentially impactful. The results are relatively complete.

Weaknesses:

Experiments require further clarification. The interpretations would be strengthened by reorganizing the experimental design.

(1) Figures 5-6 show shifts in palatability-related GC responses from recording day 1 to recording day 2. The authors propose that this drift is due to the taste experience during recording day 1, but the study, as designed, does not directly test this idea. Without a behavioral measure collected after recording day 1 intraoral exposure, it is not possible to determine whether taste preference was altered by that experience, nor whether the neural responses collected on recording day 2 represent current or most recent palatability expression vs something else. The authors' conclusion would be strengthened by adding an intervening brief access test between recording days 1 and 2. The authors could then determine whether the behavioral preferences changed after intraoral taste exposure on recording day 1, as well as whether the new set of taste-related palatability responses correlates with the updated taste preferences.

(2) The current experimental design exposes animals to 3 distinct sets of substances. These substances differ in identity (some rats never experienced sweet, while others did not experience bitter during the recording sessions) and concentration (ranging from very aversive to slightly aversive or possibly even neutral). Because palatability is known to be comparative depending on the other substances available and concentration-dependent, this introduces challenges to interpretation.

The authors state that "no differences in effects were observed between taste batteries" (Methods), but it is not clear which analyses were performed to determine the lack of difference, especially considering that many of the analyses are within-animal. Without more clarity, it is difficult to evaluate whether the interaction of different tastes within the sets of stimuli biases the main conclusions.

(3) Responses to sweet tastes are not reported in the electrophysiology data. This is seemingly the case because rats given set 1 received no sweet stimulus while rats given set 2 received 2 distinct sweet tastes. Finally, rats given set 3 did not receive quinine, yet quinine is reported in electrophysiology data.

(4) The choice of reporting average lick cluster size is problematic because the authors use thirsty rats with 10-second-long trials. Thirsty rats are likely to lick in relatively long clusters, especially for neutral and palatable tastes. If the rat is mid-cluster when the trial ends, the final cluster would be cut off prematurely, resulting in shorter overall average lick cluster size, disproportionately affecting neutral and palatable tastes over aversive tastes.

(5) Canonical palatability rankings may not apply to the concentrations selected in every stimulus set. This is particularly true for set 1, which included two concentrations of citric acid and quinine for the behavior. It is also not clear which concentrations are reported in Figures 3A2 and 3B2. Meanwhile, the concentrations of quinine and citric acid used for electrophysiology are quite low.

<https://doi.org/10.7554/eLife.110155.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

...It is unclear whether there are any systematic changes in preferences over the course of testing that could explain the observed changes in correlation with neural responses, such as changes due to learning (e.g., flavor nutrient conditioning, relief of neophobia), changes in deprivation state, or habituation to/proficiency with the BAT setup.

For the revision, we will add analysis (including either additional panels for Figure 3 or as a new Figure between what are now Figures 3 & 4) testing the hypothesis that preference

changes across testing days are non-random. Concretely, we will test: 1) whether the preference for palatable tastes increase with experience (a result that would make sense given research on neophobia; 2) whether the preference for aversive tastes decrease with experience; and 3) whether absolute consumption of any particular taste changes in a reliable direction from session to session.

A secondary point is whether any changes in preference are attributed to internal individual versus external contextual factors. Both types of variation (i.e., across individuals and across time within an individual) are mentioned in the introduction, but it is not clear what the authors believe about the nature or neural representation of these sources of variation.

While we assume that differences between rats are due to internal factors (given the controlled home-cage environment), we can't be sure that some subtle, subthreshold (for us as observers) factor impacts taste preferences. Similarly, while changes across time within an individual is categorically within the individual, we cannot be sure whether some subtle facet of their experiences determines how preferences change (as opposed to it being purely internal). We will add prose to the Discussion section on this topic—including citation of Hilary Schiff's recent work showing nurture-related preference changes as part of this new prose.

With respect to neural data analysis, no individual animal/day data are shown, making it difficult to assess the extent to which differences in correlation match individual differences in preferences and/or changes in preference with time within individuals.

The revision will include Figure panels (with analysis) showing the relationships between individual neural responses and consumption in the first and last BAT tests for 1-2 representative rats.

The correlation analysis is also lacking control for the fact that there is a certain degree of "chance" associated with behavioral and neural measures having matching ranks.

Certainly chance cannot explain our results, which consist mainly of within-rat differences in match (i.e., specific enhancement of that match for the most recent behavioral assessment)—a finding that is all the more surprising given that: 1) 2 weeks separate that behavior test and the electrophysiology session; and that 2) that 2-week gap is only 1-3 days less than the gap using the first behavioral test (that reliably correlates less well with the neural data). Nonetheless, we will add an independent, convergent analysis to the revision, testing whether the observed pattern vanishes when we shuffle the preference ranks in the behavioral data—if the result is based on chance, this shuffling should have no impact on the neural-behavioral match.

Finally, ...it is unclear to what extent changes in correlation may be attributed to overall changes in responsiveness of the neural population.

We will include a new analysis in the revision testing the hypothesis that the reduction in match between the neural and behavioral rankings reflects changes in neural excitability—spontaneous and taste-driven—between the first and second electrophysiology sessions.

Reviewer #2 (Public review):

The manuscript could use additional corollary analyses to provide a more complete picture of the phenomenon. For instance, how many neurons (per animal and in total) have significant correlations with the final BAT patterns? And with the first BAT? Can a time course of such counts be provided? Can some decoding analyses be performed at a single session level to reconstruct a rat's behavioral preference pattern from its neural activity?

These are all really good ideas. We are in the process of implementing all but the last; we will attempt the last as well, but can't promise that we have large enough ensembles to provide stable results of such a subtle decoding task (reflecting the last BAT session's preference pattern significantly better than the first session's pattern).

The manuscript could benefit from additional polishing, both in the text as well as in the figures.

It is being done, on the basis of suggestions made by R2 in the non-public comments.

Reviewer #3 (Public review):

Without a behavioral measure collected after recording day 1 intraoral exposure, it is not possible to determine whether taste preference was altered by that experience...The authors' conclusion would be strengthened by adding an intervening brief access test between recording days 1 and 2.

We very much appreciate Reviewer 3's suggestion, but the primary authors involved in data collection on this project have moved on, and we won't be able to collect the additional dataset that would be required. Instead, we will soften the conclusion that we reach in the last section, and suggest this experiment as a future direction.

The current experimental design exposes animals to 3 distinct sets of substances ... [that] differ in identity ... and concentration. Because palatability is known to be comparative depending on the other substances available and concentration-dependent, this introduces challenges to interpretation, [and] without more clarity, it is difficult to evaluate whether the interaction of different tastes within the sets of stimuli biases the main conclusions.

This is an interesting point. We hope that some of the work that we are undertaking in response to Reviewers 1 & 2 (see above) will shed light on whether there is any non-randomness in between-session preference changes; such non-randomness would imply that we might want to conclude that preferences change more with one battery than another. But we will perform a more direct test of this hypothesis, breaking the dataset apart and asking whether our phenomena are observed more with one battery than another. If it turns out that the magnitude of the impact of experience does depend on the nature of the taste battery (we predict not, for reasons that are in the manuscript), we shall introduce that complexity into our interpretation, and the Discussion thereof.

Responses to sweet tastes are not reported in the electrophysiology data. This is seemingly the case because rats given set 1 received no sweet stimulus while rats given set 2 received to 2 distinct sweet tastes. Finally, rats given set 3 did not receive quinine, yet quinine is reported in electrophysiology data.

We are unsure of the source of this confusion—in every case, the rat received the same tastes in the electrophysiology sessions that were delivered in the BAT preference tests—but we will modify the text to ensure: 1) that panels reflecting data from a single rat (panels that will therefore necessarily include only a subset of possible tastes) are clearly marked as such; and 2) that the nature of which taste batteries were delivered is more explicit.

The choice of reporting average lick cluster size is problematic because the authors use thirsty rats with 10-second-long trials. Thirsty rats are likely to lick in relatively long clusters, especially for neutral and palatable tastes. If the rat is mid-cluster when the trial ends, the final cluster would be cut off prematurely, resulting in shorter overall average lick cluster size, disproportionately affecting neutral and palatable tastes over aversive tastes.

We have ourselves been deeply concerned with this issue; we have recently published a paper that includes within it a direct test demonstrating that calculations of lick bout lengths from 10-sec BAT trials result in taste palatability estimates that are identical to (and less noisy than) those generated from more classically-used 15-min ad lib licking. We will cite this paper (Lin, et al., 2026) in the Methods section of the revision, along with text clarifying how we calculated lick clusters. That said, we are also planning to conduct an additional analysis that estimates taste preference after removing these “premature bouts” and will evaluate how this recalculation affects our results.

Of course, even if 10-sec BAT trial data DIDN'T provide reliable preference measures, the result of clusters being cut short by the end of a trial would be an underestimation of the preference for the palatable tastes (which drive far more licking than aversive tastes and are therefore more likely to be mid-bout at the end of a trial). Such an underestimation would in turn be expected to reduce the observed neural-behavioral correlation. This fact actually highlights the robustness of our findings.

Canonical palatability rankings may not apply to the concentrations selected in every stimulus set. This is particularly true for set 1, which included two concentrations of citric acid and quinine for the behavior. It is also not clear which concentrations are reported in Figures 3A2 and 3B2. Meanwhile, the concentrations of quinine and citric acid used for electrophysiology are quite low.

In the revision Methods section, we will explicitly motivate our reasoning behind canonical rankings for each taste battery used (the added text will include citations). We have also added to the Discussion section prose concerning the possible impact of possibly getting those rankings wrong—i.e., the impact is minimal, given that our results are largely driven by differences between rats (and day-to-day differences within rat), and the resultant fact that almost any choice of canonical rankings would poorly reflect the behavior of individual rats on individual days.

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